



QU Chat Bot

Automated Knowledge Base & AI Assistant

Course: CS498 - Graduation Project

Supervisor: Dr. Abdulgader Al-Maymuni

Academic Year: 1447/1448 (2025/2026)

Team Members

- **Suliman S. Al-Ghofaili**
421109766 • Team Lead, Backend/AI
- **Rayan N. Al-Harbi**
421109529 • Frontend/UI
- **Khalifah S. Al-Khalifah**
431109692 • Data Pipeline
- **Faris I. Al-Awaji**
431109338 • Data Pipeline
- **Sultan M. Al-Mutairi**
421109632 • Data Pipeline
- **Abdulrahman H. Al-Resheed**
421107788 • Frontend/UI

May 2026

PRESENTATION OUTLINE

Agenda

A comprehensive overview of our graduation project journey

 Estimated Duration: 20 minutes

01



Problem Statement

Challenges & Pain Points

02



Proposed Solution

AI-Powered Assistant

03



System Architecture

Technical Design

04



Tech Stack

Technologies Used

05



Key Features

Core Capabilities

06



Testing & Results

Performance Metrics

07



Vision 2030

National Alignment

08



Future Work

Enhancements & Growth

THE CHALLENGE

Problem Statement

Current university support systems face critical limitations that hinder student success

 4 Key Issues Identified



Information Fragmentation

University information scattered across multiple disconnected sources and platforms



Outdated Information

Inconsistent and outdated data across different university systems and channels



Support Bottlenecks

Staff overwhelmed with repetitive FAQs, creating long wait times for students



Time Wastage

Students spend excessive time searching instead of focusing on studies

OUR INNOVATION

Proposed Solution

An AI-powered chatbot system that provides instant, accurate, and bilingual support to QU students

✦✦ 4 Core Technologies



Bilingual AI Assistant

Arabic & English support with natural RTL processing

AR

EN



RAG Architecture

Retrieval-Augmented Generation for grounded, accurate answers

Context-Aware



GPT-4o-mini & Pinecone

Powered by OpenAI and vector database for fast semantic search

8,577 Vectors



n8n Automation

Self-updating knowledge base from official university sources

Auto-Refresh

PROJECT GOALS

Objectives & Aims

Strategic goals driving our AI assistant development for QU students

4 Core Objectives



24/7 AI Assistant

Deploy round-the-clock RAG-powered support system for QU students with instant responses

Always Available



Automated Updates

Self-updating knowledge base from official QU sources using n8n workflows

Auto-Sync



Bilingual Chat

Real-time bilingual experience with Arabic & English, including RTL support

AR + EN



Containerized

Scalable architecture with Docker containerization for reliable deployment

Scalable

TECHNICAL DESIGN

System Architecture

End-to-end flow from user interaction to AI-powered response generation

 4 Core Components



Next.js Frontend

React-based UI with
TypeScript, Tailwind CSS, and
shadcn/ui components



n8n Webhook

Workflow automation platform
for orchestrating AI agent and
data flow



Request Flow



AI Agent

GPT-4o-mini powered chatbot
with RAG architecture for
grounded answers



Pinecone Vector DB

8,577 vector embeddings for
semantic search and context
retrieval

TECHNOLOGIES

Tech Stack

Modern technologies powering our AI assistant system

 5 Categories



Frontend

Next.js 16

TypeScript

Tailwind CSS

shadcn/ui

Framer Motion



Backend

n8n

OpenAI GPT-4o-mini

Workflow Automation



Database

Pinecone

Vector Database

8,577 Vectors



Data Pipeline

Python

OpenAI Embeddings

423 Documents



Deployment

Vercel

n8n Cloud

CI/CD Pipeline

99% Uptime Target

CAPABILITIES

Key Features

Powerful features designed for seamless student support experience

✦✦✦ 6 Core Features



Bilingual Support

Arabic & English with full RTL support for natural interaction



RAG Architecture

Grounded, accurate answers backed by verified knowledge base



Fast Response

< 3 seconds average response time for instant support



Modern UI

Responsive design with Dark/Light themes for better experience



Message Rating

User feedback system to continuously improve answer quality



Quick Links

Direct access to university services and important resources

DATA METRICS

Knowledge Base Stats

Comprehensive knowledge base powering accurate AI responses

 6 Key Metrics



423

Documents Indexed



8,577

Vector Embeddings



17

Colleges Covered



6

Service Categories



2

Languages Supported



<3s

Response Time



PROCESS FLOW

RAG Pipeline

Retrieval-Augmented Generation workflow
for accurate, grounded answers

 5 Processing Steps



User Question

1

Student submits query via chat interface



Query Embedding

2

Generate vector representation using OpenAI



Vector Search

3

Find top-5 similar documents in Pinecone DB



AI Processing

4

Context + question sent to GPT-4o-mini



Grounded Answer

5

Verified response returned to student with source citations and high confidence

User Interface

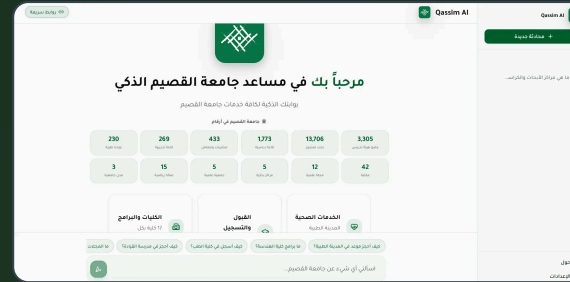
INTERFACE DESIGN

Modern, responsive design optimized for student experience

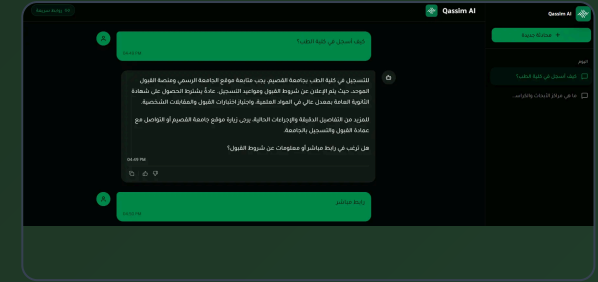
4 Key Screens



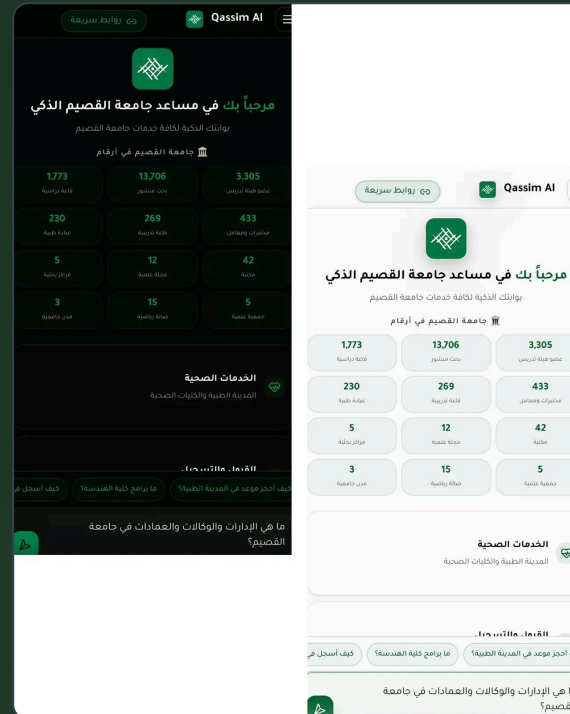
Welcome Screen



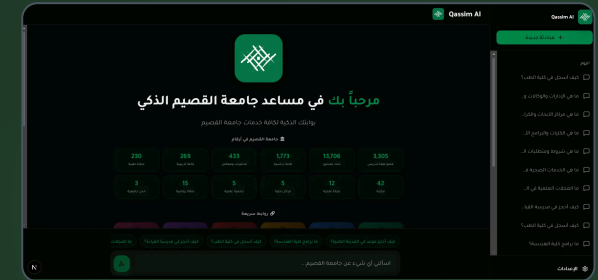
Chat Interaction



Mobile Responsive



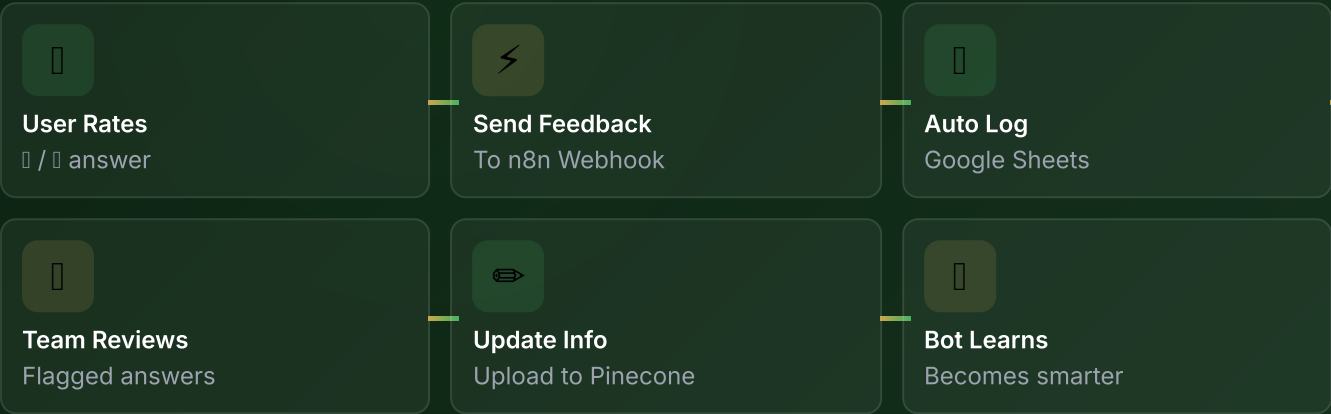
Dark Mode



FEEDBACK MECHANISM

Continuous Improvement System (Human-in-the-Loop)

A feedback mechanism that continuously improves the chatbot's accuracy by learning from real user interactions.

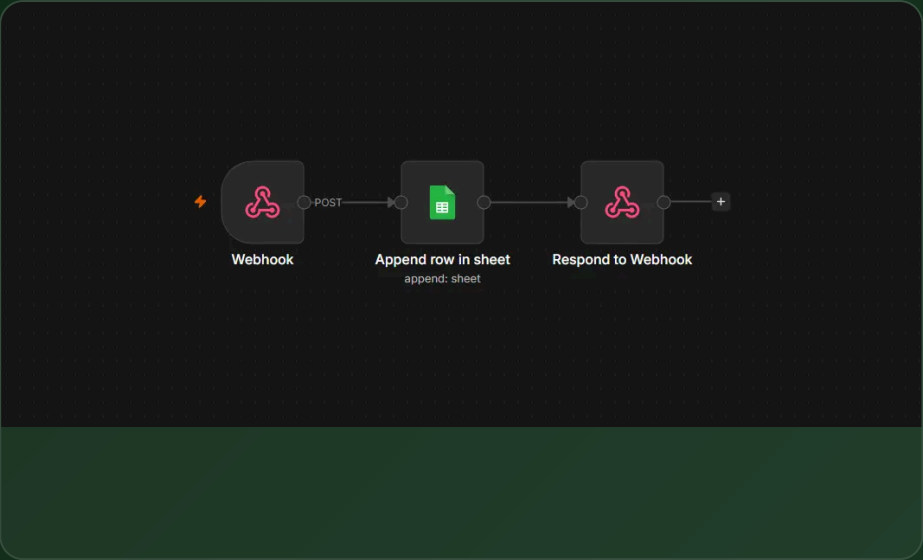


What Gets Logged

- Timestamp
- Question
- Bot's Answer
- Rating
- Language
- Status

Key Benefit

Every thumbs-down response is automatically captured, allowing the team to identify knowledge gaps and refine the bot over time — no manual tracking needed.



n8n Feedback Logger Workflow

Timestamp	Question	Answer	Rating	Language	Status	Notes
10:00:00 2024/05/29	كيف يمكنني التسجيل في جامعة القصيم؟	يمكنك التسجيل عبر بوابة القبول الإلكترونية في موقع الجامعة الرئيسي.	Good	Arabic	Fixed	تم توجيه المستخدم إلى البوابة الصحيحة.
10:05:00 2024/05/29	What are the library hours?	The library is open from 8 AM to 9 PM on weekdays.	Bad	English	Pending	Answer needs more detail about weekend hours.
10:10:00 2024/05/29	أين يقع مبنى كلية الهندسة؟	يقع مبنى كلية الهندسة في الجهة الغربية من المدينة الجامعية.	Good	Arabic	Fixed	الموقع دقيق.
10:15:00 2024/05/29	How to reset my password?	You can reset your password through the IT self-service portal.	Bad	English	Ignored	User mentioned the link was broken.
10:20:00 2024/05/29	متى موعد الاختبارات النهائية؟	نبدأ الاختبارات النهائية في تاريخ 15 يونيو حسب التقويم الأكاديمي.	Good	Arabic	Fixed	تم التأكد من المواعيد.
2026-05-29T14:05:08.657	تم إرسال رسالة قبول كلية الهندسة	هذه المعلومات غير معروفة بشكل محدد في قاعدة بيانات الجامعة التي نحصل عليها. يرجى التواصل مع عمادة القبول والتسجيل في الجامعة عبر البريد الإلكتروني 0163800050 أو زيارة موقع الجامعة الرئيسي qu.edu.sa.	Bad	Arabic	Pending	

Live Feedback in Google Sheets

PLATFORM SELECTION

Why n8n?

Superior workflow automation platform for
AI integration

★ 9.5/10 Overall Score



Open-Source

- ✓ Transparent codebase, free to use & modify
- ✗ Zapier: Paid subscription required



Self-Hostable

- ✓ Full control over data & deployment
- ✗ Others: Cloud-only, vendor lock-in



Visual Builder

- ✓ Intuitive drag-and-drop interface
- ✗ Make: Complex learning curve



AI Integration

- ✓ Built-in OpenAI & Pinecone nodes
- ✗ Others: Limited or no AI support



Docker Support

Containerized deployment for scalability and portability

9.5/10

Overall Rating

VALIDATION

Testing & Results

Comprehensive testing validates system performance and reliability

 Performance Metrics



Response Time

<3s

95th percentile latency



Uptime

99%

Reliability target



RAG Quality Metrics

0.89

Context Precision

0.92

Faithfulness

0.95

Answer Relevance



Real Test Scenarios

-  Admissions inquiries - accurate procedural guidance
-  Course registration - step-by-step walkthrough
-  Scholarship queries - verified source citations

NATIONAL VISION

Vision 2030

Aligning with Saudi Arabia's strategic goals
for digital transformation

3 Key Programs



Digital Transformation

- ✓ AI-powered services
- ✓ Smart automation
- ✓ 24/7 digital access



Human Capacity

- ✓ Student empowerment
- ✓ Knowledge accessibility
- ✓ Skill development



Quality of Life

- ✓ Reduced wait times
- ✓ Better services
- ✓ Enhanced experience



National Impact

Contributing to Saudi Arabia's digital future

3

Programs

2030

Target Year

PROJECT OUTCOMES

Conclusion & Future Work

Successfully deployed an AI-powered assistant for QU students

 Ready for Production



Key Achievements



24/7 AI support system deployed



Bilingual responses with high accuracy



Self-updating knowledge base



RAG architecture for grounded answers



Future Enhancements



SIS Integration

Personalized student information access



Course Catalog

Real-time course information & schedules



Analytics Dashboard

Insights into student queries & trends



Multi-modal Support

Images & PDF processing capabilities



Live Demo

Access the deployed application

qu-smart-assistant.vercel.app



Thank You

We appreciate your attention and welcome your questions

 Questions & Answers



Questions?

We're happy to answer your inquiries



Live Demo

qu-smart-assistant.vercel.app



GitHub

github.com/9llmy/qu-smart-assistant



Project Team

 Suliman S. Al-Ghofaili

 Rayan N. Al-Harbi

 Khalifah S. Al-Khalifah

 Faris I. Al-Awaji

 Sultan M. Al-Mutairi

 Abdulrahman H. Al-Resheed